

Stochastic Mathematics for Finance

Week 3 Assignment

Kamana Gupta
240509

1 Question 1: Brownian Motion Paths and Quadratic Variation

(a) Four Properties of Standard Brownian Motion

A standard Brownian motion $\{W_t\}_{t \geq 0}$ satisfies:

1. $W_0 = 0$ almost surely.

Financial interpretation: No uncertainty has accumulated at time 0.

2. Independent increments:

$$W_t - W_s$$

is independent of the past for $0 \leq s < t$.

Financial interpretation: Future price movements do not depend on past movements.

3. Stationary Gaussian increments:

$$W_t - W_s \sim N(0, t - s).$$

Financial interpretation: Returns over equal time intervals have the same distribution.

4. Continuous sample paths.

Financial interpretation: Prices evolve continuously without sudden jumps.

(b) Showing that $\text{Cov}(W_s, W_t) = \min(s, t)$

Assume $s < t$. Then

$$W_t = W_s + (W_t - W_s).$$

Therefore,

$$\text{Cov}(W_s, W_t) = \text{Cov}(W_s, W_s) + \text{Cov}(W_s, W_t - W_s).$$

Since $W_t - W_s$ is independent of W_s ,

$$\text{Cov}(W_s, W_t - W_s) = 0.$$

Also,

$$\text{Var}(W_s) = s.$$

Hence,

$$\text{Cov}(W_s, W_t) = s.$$

Therefore,

$$\boxed{\text{Cov}(W_s, W_t) = \min(s, t)}.$$

(c) Expected Value of Quadratic Variation

For a partition

$$0 = t_0 < t_1 < \cdots < t_n = T,$$

the quadratic variation is

$$Q_n = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2.$$

Since

$$W_{t_{k+1}} - W_{t_k} \sim N(0, \Delta t_k),$$

we have

$$E[(W_{t_{k+1}} - W_{t_k})^2] = \Delta t_k.$$

Thus,

$$E[Q_n] = \sum_{k=0}^{n-1} \Delta t_k = T.$$

Hence,

$$\boxed{E[Q_n] = T}.$$

(d) Convergence of Q_n

For a normal random variable $Z \sim N(0, \sigma^2)$,

$$\text{Var}(Z^2) = 2\sigma^4.$$

Therefore,

$$\text{Var}(Q_n) = \sum_{k=0}^{n-1} 2(\Delta t_k)^2.$$

Let

$$\|\Pi\| = \max_k \Delta t_k.$$

Then

$$\text{Var}(Q_n) \leq 2\|\Pi\| \sum_{k=0}^{n-1} \Delta t_k = 2T\|\Pi\|.$$

As $\|\Pi\| \rightarrow 0$,

$$\text{Var}(Q_n) \rightarrow 0.$$

Since $E[Q_n] = T$,

$$E[(Q_n - T)^2] = \text{Var}(Q_n) \rightarrow 0.$$

Hence,

$$\boxed{Q_n \rightarrow T \text{ in } L^2.}$$

Although the Brownian path is random, its quadratic variation converges to the deterministic value T . This property is fundamental in Itô calculus and distinguishes Brownian motion from ordinary differentiable functions.

(e) Brownian Motion Path and Quadratic Variation

A Brownian motion path was simulated on $[0, 1]$ using $N = 10,000$ steps.

Table 1: Single Path Statistics

Quantity	Value
Empirical QV	1.0068
Theoretical QV	1.0000
Total Variation	79.95

The quadratic variation is close to 1, while the total variation is much larger.

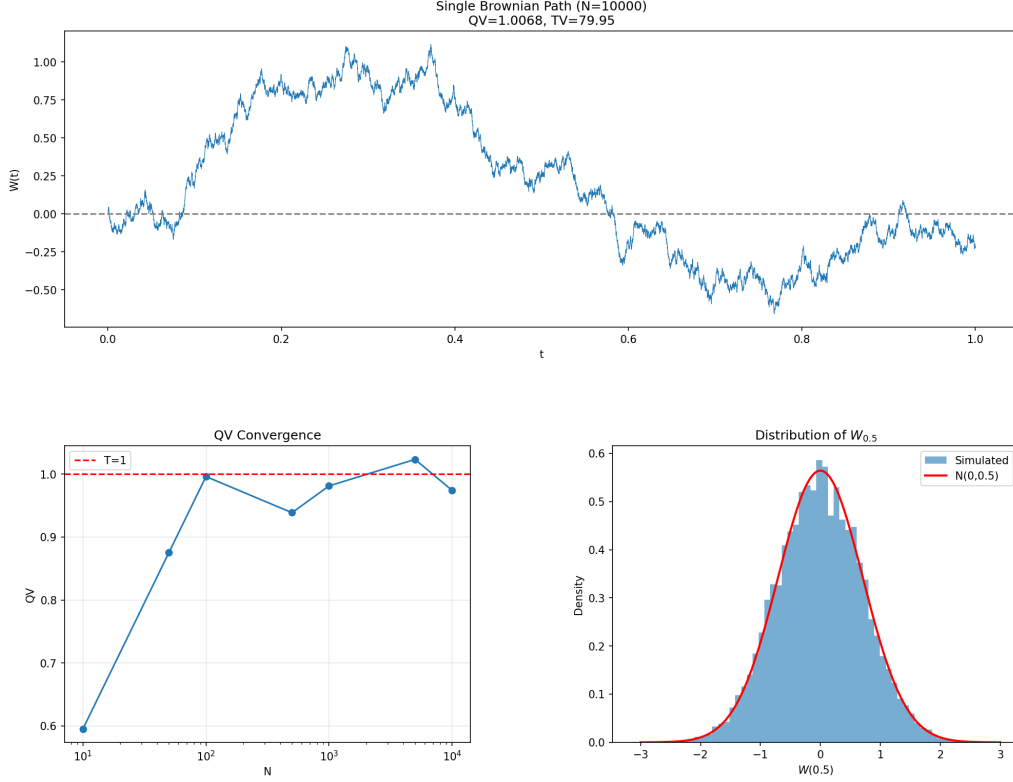


Figure 1: Brownian motion path, quadratic variation convergence, and distribution of $W_{0.5}$.

(f) Convergence of Quadratic Variation

The estimated quadratic variation approaches the theoretical value $T = 1$ as the partition becomes finer.

$$Q_n \rightarrow 1.$$

For small values of N , the estimates fluctuate noticeably. As N increases, the empirical quadratic variation becomes more stable and remains close to the theoretical value. Figure ?? shows the convergence of the empirical quadratic variation as the number of partition points increases.

(g) Distribution of $W_{0.5}$

Using 10,000 simulated values:

Table 2: Distribution of $W_{0.5}$	
Statistic	Value
Empirical Mean	-0.012
Theoretical Mean	0
Empirical Variance	0.514
Theoretical Variance	0.500

The empirical values are close to the theoretical values of a normal distribution:

$$W_{0.5} \sim N(0, 0.5).$$

2 Question 2: Itô's Lemma and Geometric Brownian Motion

(a) Itô's Lemma

Let

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t.$$

For a twice continuously differentiable function $f(t, x)$,

$$df(t, X_t) = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x dW_t.$$

The additional term

$$\frac{1}{2} \sigma^2 f_{xx} dt$$

appears because

$$(dW_t)^2 = dt.$$

Since Brownian motion has non-zero quadratic variation, second-order terms cannot be ignored as in ordinary calculus.

(b) Applying Itô's Lemma to $f(S_t) = \ln(S_t)$

For the GBM

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

take

$$f(S_t) = \ln(S_t).$$

Then

$$f_x = \frac{1}{S_t}, \quad f_{xx} = -\frac{1}{S_t^2}.$$

Applying Itô's Lemma,

$$d(\ln S_t) = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t.$$

Integrating from 0 to t ,

$$\ln \left(\frac{S_t}{S_0} \right) = \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t.$$

Hence,

$$S_t = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right].$$

(c) Itô Correction

Using the GBM solution,

$$E[S_t] = S_0 e^{\mu t}.$$

For

$$S_0 = 100, \quad \mu = 0.07, \quad \sigma = 0.2, \quad t = 10,$$

Table 3: Effect of Omitting the Itô Correction

Quantity	Value
Correct $E[S_{10}]$	201.38
Without correction	245.96
Relative Error	22.1%

Ignoring the correction overestimates the expected stock price.

(d) Euler–Maruyama Simulation

The Euler–Maruyama scheme used was

$$S_{k+1} = S_k + \mu S_k \Delta t + \sigma S_k \Delta W_k.$$

A total of 50,000 paths were simulated with

$$S_0 = 100, \quad \mu = 0.07, \quad \sigma = 0.2, \quad T = 1.$$

(e) Terminal Stock Price Distribution

Table 4: Terminal Stock Price Statistics

Statistic	Empirical	Theoretical
Mean	107.20	107.25
Standard Deviation	21.49	21.67

The empirical values are close to the theoretical values.

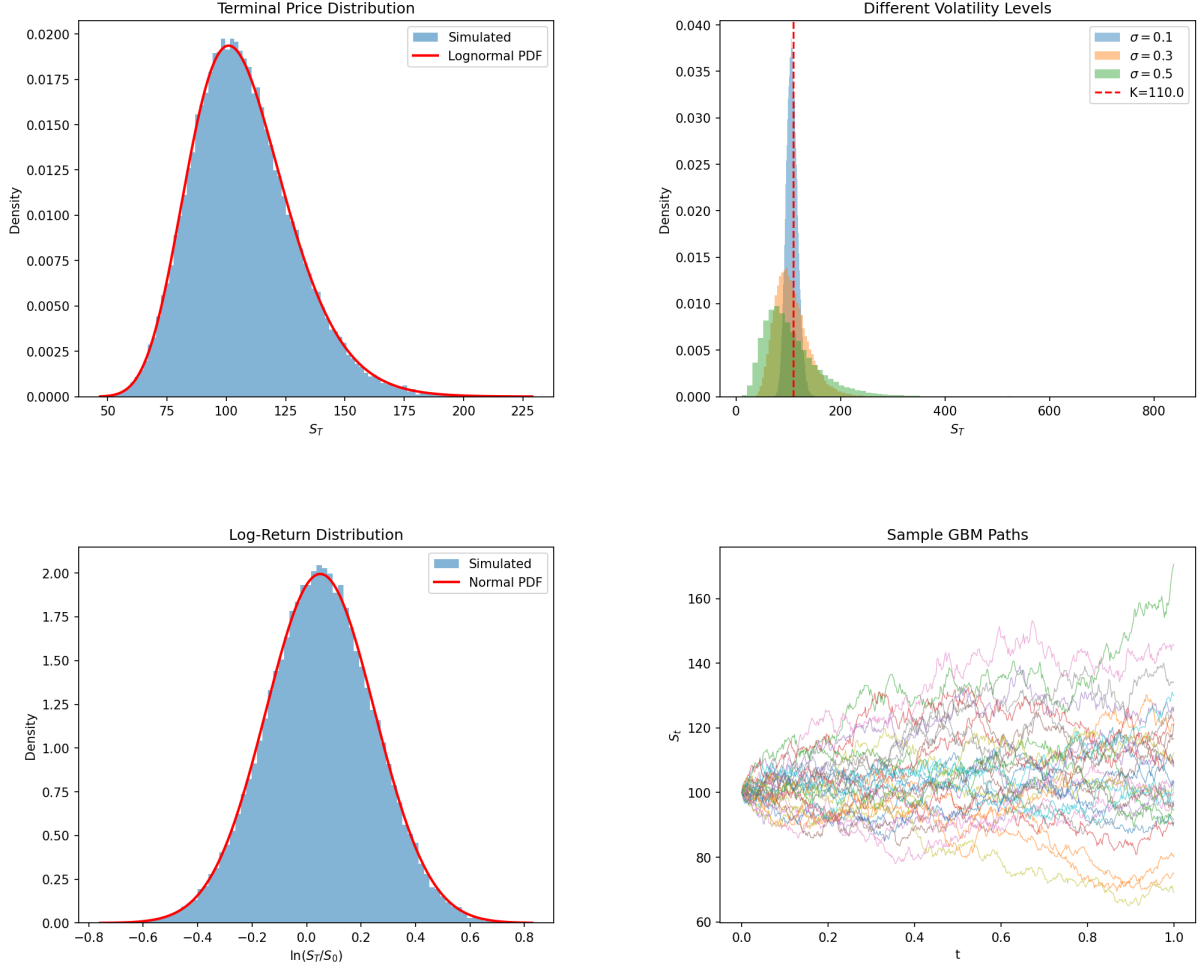


Figure 2: Terminal stock price distribution, volatility comparison, log-return distribution and sample GBM paths.

The figure shows the terminal stock price distribution, the effect of volatility, the distribution of log-returns, and sample GBM paths.

(f) Effect of Volatility

Table 5: Results for Different Values of σ

σ	$E[S_T]$	$\text{Std}(S_T)$	$E[(S_T - 110)^+]$
0.1	107.20	10.75	3.07
0.3	106.74	32.62	11.32
0.5	107.72	57.70	20.39

As σ increases, the spread of S_T becomes larger. The expected call payoff also increases.

(g) Distribution of Log>Returns

For GBM,

$$\ln \left(\frac{S_T}{S_0} \right) \sim N \left(\left(\mu - \frac{\sigma^2}{2} \right) T, \sigma^2 T \right).$$

The simulation results are:

Table 6: Log-Return Statistics		
Statistic	Empirical	Theoretical
Mean	0.0498	0.0500
Variance	0.0395	0.0400

The histogram is consistent with the theoretical normal distribution.

3 Question 3: Euler–Maruyama and the Ornstein–Uhlenbeck Process

(a) Drift and Diffusion Coefficients

The Ornstein–Uhlenbeck process is

$$dX_t = -\theta X_t dt + \sigma dW_t.$$

Hence,

$$a(t, X_t) = -\theta X_t$$

and

$$b(t, X_t) = \sigma.$$

The drift term pulls the process towards zero. Larger values of θ lead to faster mean reversion.

(b) Euler–Maruyama Discretisation

Applying Euler–Maruyama,

$$X_{k+1} = X_k - \theta X_k \Delta t + \sigma \Delta W_k.$$

Equivalently,

$$X_{k+1} = (1 - \theta \Delta t) X_k + \sigma \Delta W_k.$$

For stability,

$$|1 - \theta \Delta t| < 1.$$

Therefore,

$$\Delta t < \frac{2}{\theta}.$$

For

$$\theta = 2,$$

the condition becomes

$$\Delta t < 1.$$

Since

$$\Delta t = 0.001,$$

the scheme is stable.

(c) Stationary Distribution

The stationary distribution of the OU process is

$$X_\infty \sim N\left(0, \frac{\sigma^2}{2\theta}\right).$$

Using

$$\sigma = 0.5, \quad \theta = 2,$$

$$\frac{\sigma^2}{2\theta} = \frac{0.25}{4} = 0.0625.$$

Thus,

$$X_\infty \sim N(0, 0.0625).$$

As time increases, the distribution of X_t approaches this stationary distribution regardless of the starting value. This is useful in interest rate modelling because the process remains centred around a long-run level instead of drifting indefinitely.

(d) Simulated Paths

Ten sample paths were generated with

$$X_0 = 3, \quad \theta = 2, \quad \sigma = 0.5, \quad T = 5.$$

The paths move toward zero and then fluctuate around the stationary mean.

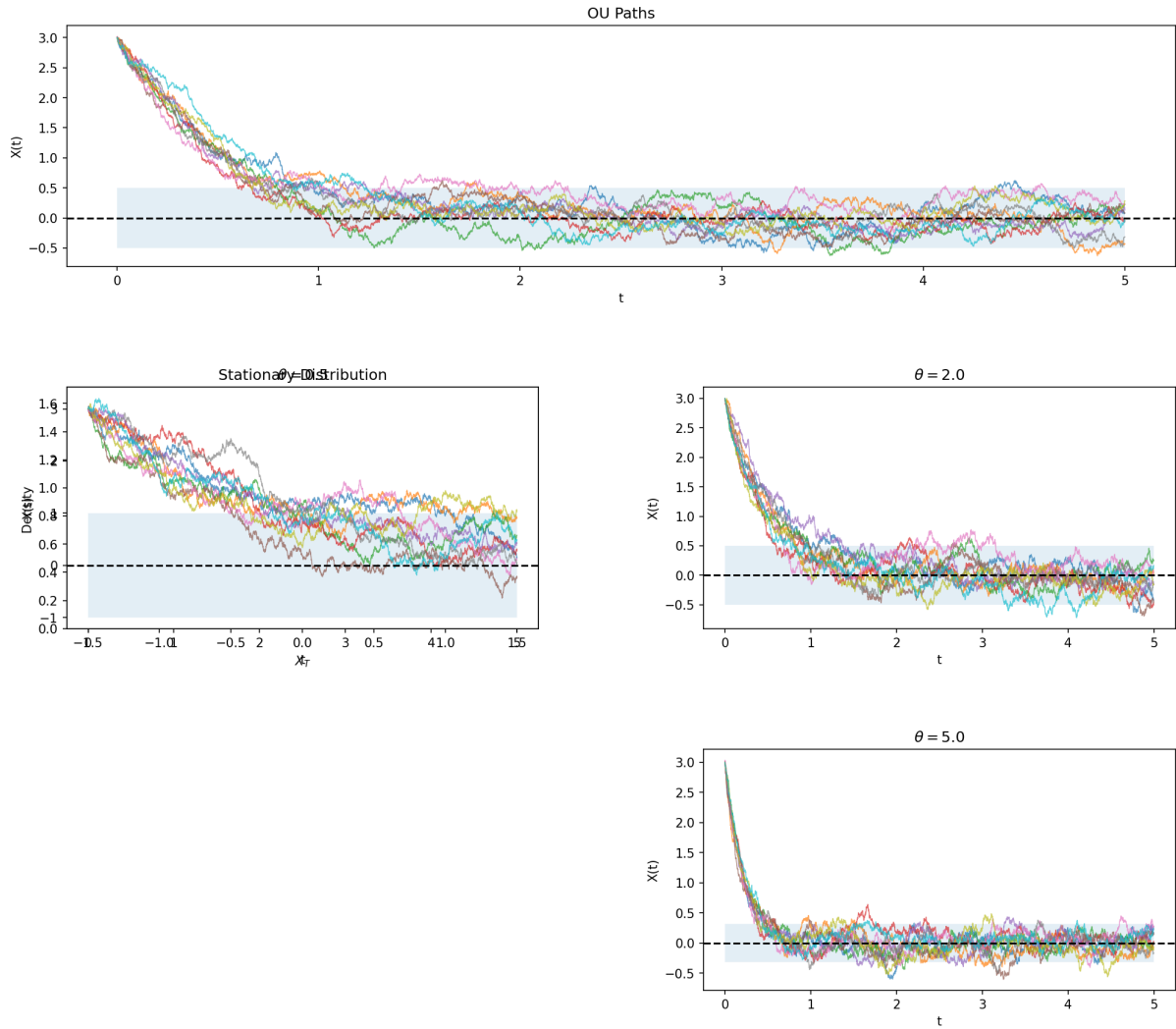


Figure 3: Numerical results for the Ornstein–Uhlenbeck process.

The figure shows the mean-reverting behaviour of the OU process and the effect of different values of θ .

(e) Stationary Distribution Verification

Using 20,000 simulated paths at $T = 10$:

Table 7: Terminal Distribution Statistics

Statistic	Empirical	Theoretical
Mean	-0.005	0
Variance	0.0625	0.0625

The empirical distribution agrees closely with the stationary distribution.

(f) Effect of Mean Reversion

For different values of θ :

Table 8: Effect of θ		
θ	$\sigma_\infty = \frac{\sigma}{\sqrt{2\theta}}$	Observation
0.5	0.500	Slow mean reversion
2.0	0.250	Moderate mean reversion
5.0	0.158	Fast mean reversion

As θ increases, the process returns to zero more quickly and the stationary spread decreases.

(g) Strong and Weak Convergence

Strong convergence measures the pathwise error:

$$E \left[|X_T^{(h)} - X_T| \right] = O(h^{1/2}).$$

Weak convergence measures the error in expectations:

$$\left| E[g(X_T^{(h)})] - E[g(X_T)] \right| = O(h).$$

For option pricing and expectation-based quantities, weak convergence is usually sufficient because only the distribution of the terminal value is required.